

Bartr

The laundromat on Murray Avenue is worth \$570,768, give or take 22 percent. Today you can buy 20 shares of it.

PITTSBURGH, PRICED TODAY	OUR VALUE
Squirrel Hill Wash and Fold	\$570,768 ±22%
Butler Street Laundromat	\$565,138 ±14%
Steel City Express Car Wash	\$1,534,521 ±37%
Iron City Precision Machine	\$979,576 ±36%
Bloomfield Coin Laundry	\$192,254 ±82%

36.2 million small businesses in America.

SBA Office of Advocacy, 2025. 99.9 percent of all businesses in the country.

80%
**of the ones that try to sell never find a
buyer.**

They shut down. Most of them are profitable.

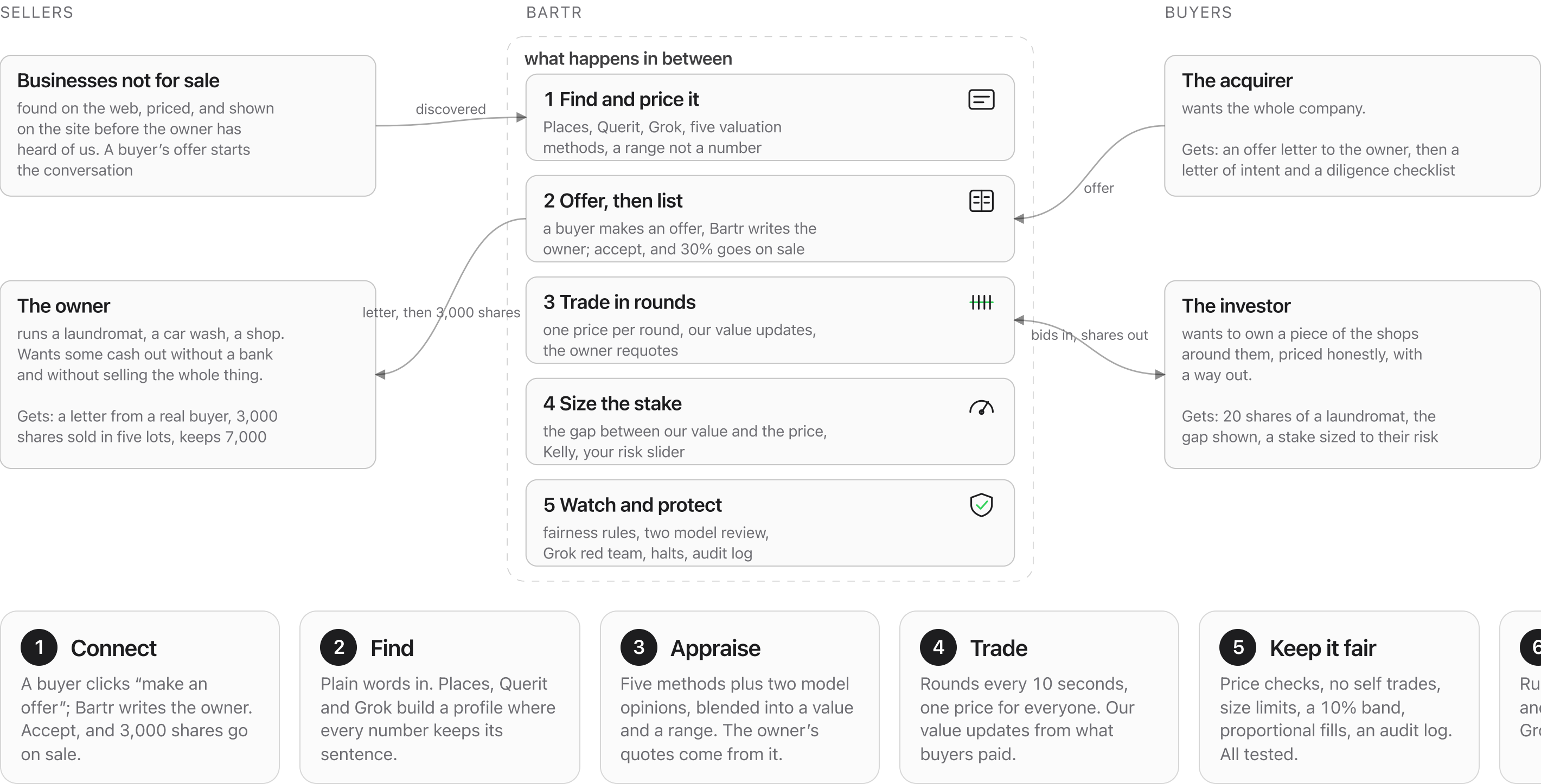
Exit Planning Institute: 20 to 30 percent of businesses that go to market sell. BizBuySell's own close rate is under 7 percent.

Half of the American workforce works for one.

46 percent of private sector jobs, and 9 of every 10 net new jobs last year. SBA Office of Advocacy, 2025.

Who Bartr is for, and what it does

We connect buyers and sellers.



Discover

Bartr

DiscoverPortfolioSurveillanceAgent

LOGINSTART TRADING

DISCOVER

laundromat in pittsburgh

SEARCH

QUERY

laundromat in pittsburgh

CAT laundromat

NAICS —

STATE PA

STATE

Any

VALUE

Any

SORT

Relevance

3 PRICED · 0 READING

COMPANY	VALUE	BID	ASK	LAST	CONFIDENCE	COMPLETE
Bloomfield Coin Laundry Pittsburgh, PA · laundromat · 4.0 (27)	\$192K	38.86	22.71	25.55		18%
Butler Street Laundromat Pittsburgh, PA · laundromat · 4.4 (118)	\$565K	68.21	58.93	58.91		86%
Squirrel Hill Wash and Fold Pittsburgh, PA · laundromat · 4.6 (214)	\$571K	60.99	57.22	57.22		78%

Bartr

A discovery engine and exchange for the businesses that will never be listed.

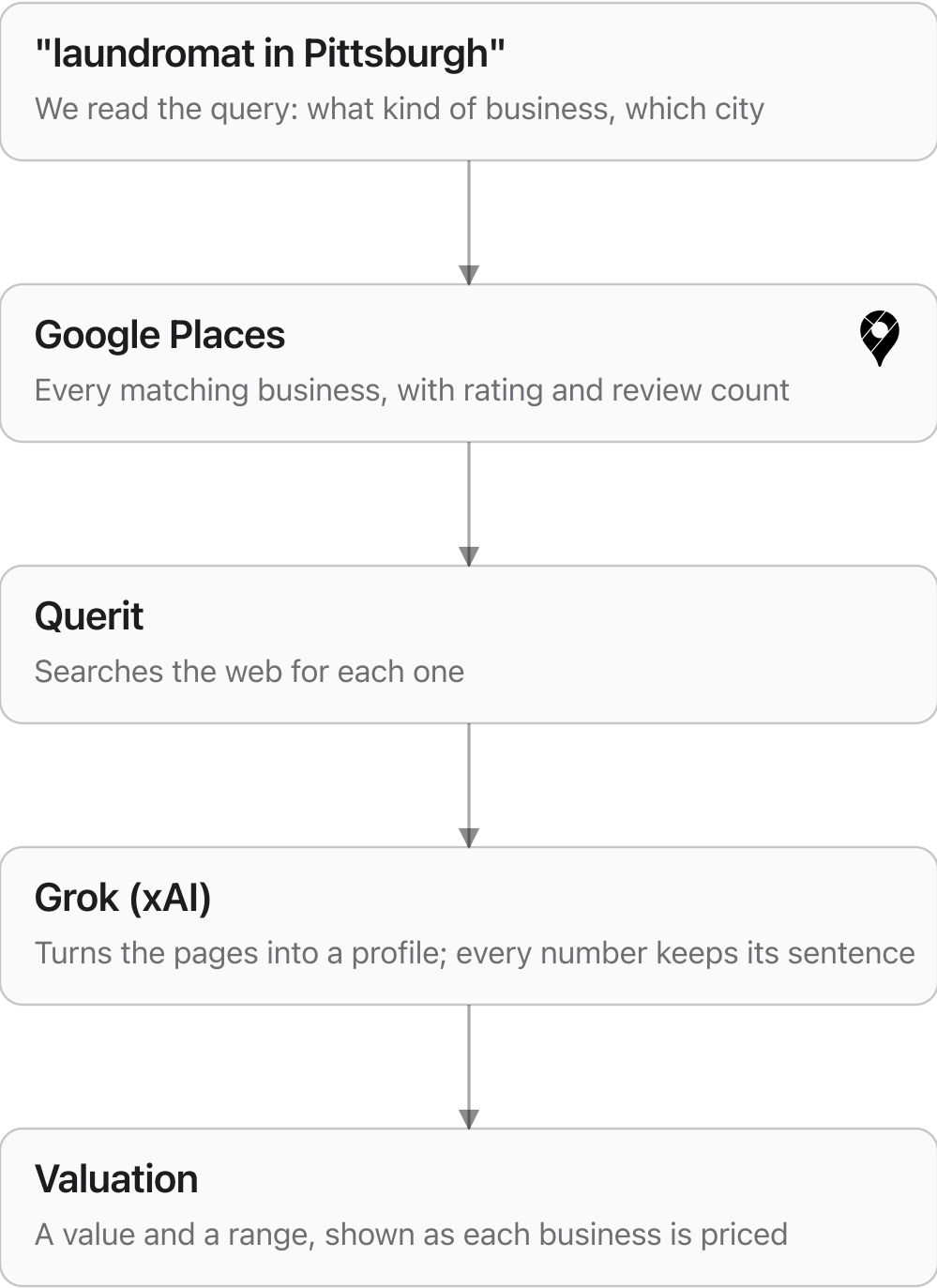
PRODUCT

Discover

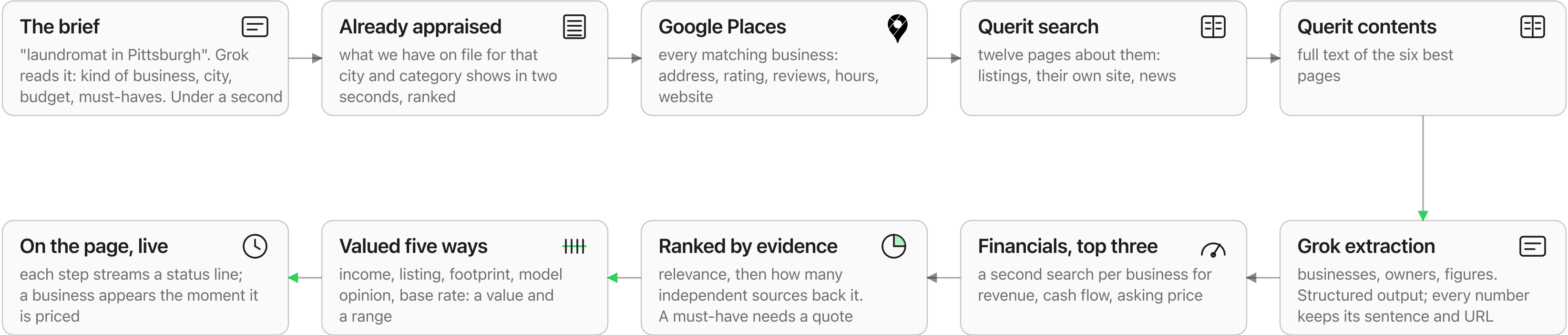
Portfolio

Surveillance

"laundromat in Pittsburgh"



Our proprietary search helps you discover underground, undervalued companies.



Google Places Text Search

Every matching business in the city: name, address, rating, review count, hours, website. This is the list that exists nowhere else in one place.

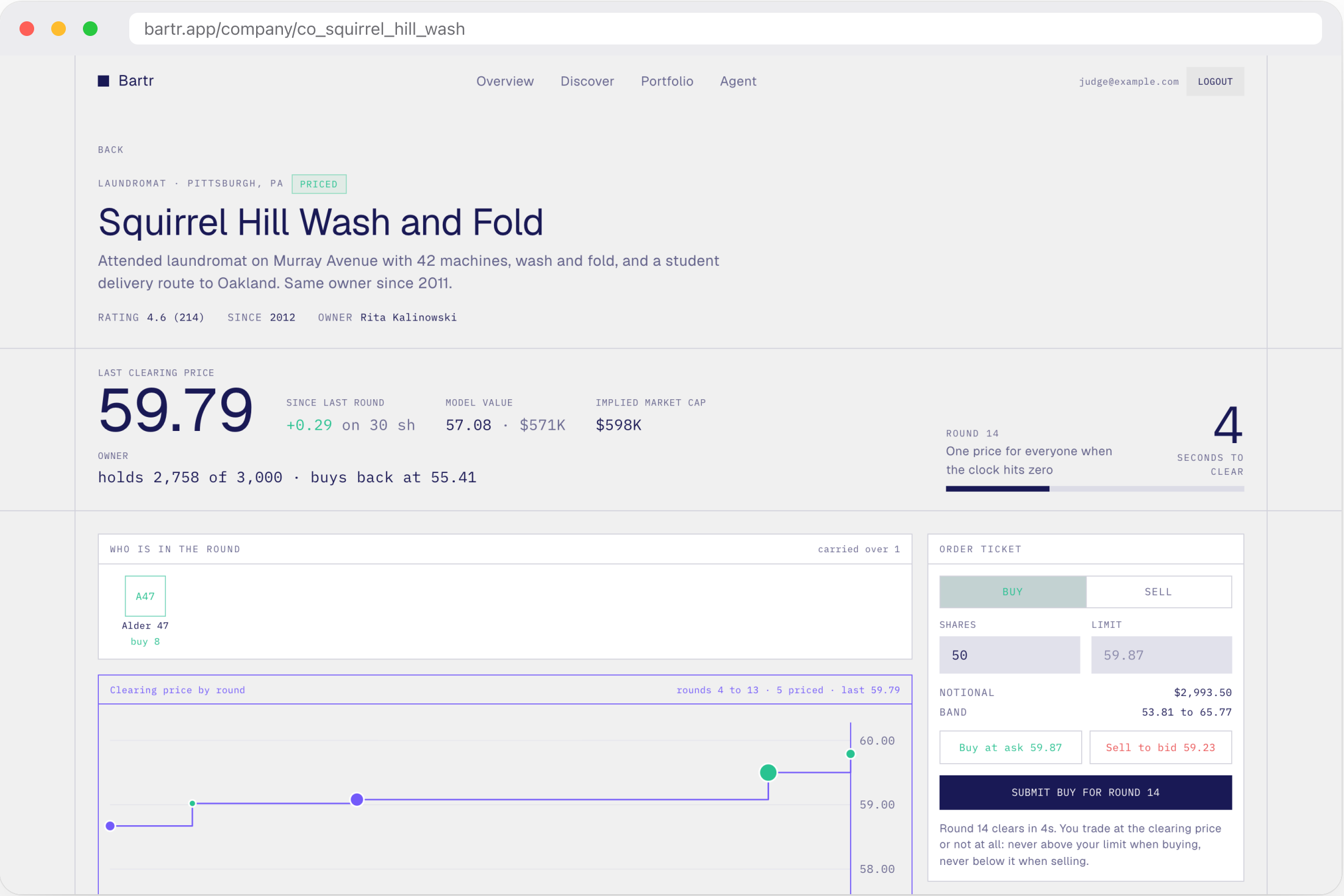
Querit search and contents

Twelve pages per query, then the full text of the best six. Listings, the business’s own site, news, reviews. A targeted second search for the top three looks for revenue, cash flow and an asking price.

Grok, twice

A fast model reads the brief in under a second. A stronger one turns pages into profiles with structured output: every figure keeps the sentence it came from and its URL. Pages are evidence, never instructions.

One company



Squirrel Hill Wash and Fold

\$570,768

Range \$475k to \$686k. We could be about 22 percent off either way.

\$57.08 a share. 10,000 shares.

On the page: the last price, every open order, and the seconds until the next round.

How we price it

and how far off we might be

Five ways to value a business. Each one comes with its own typical miss.

1. What one method says	$\ln V_k = \mu_k \pm \sigma_k$	The income method says Squirrel Hill is worth \$589k, and that method is usually off by about 25 percent. We work in logs so 2x too high and 2x too low count as the same size miss.
2. Combining five answers	$\mu = \frac{\sum_k w_k \mu_k}{\sum_k w_k}, \quad w_k = \frac{1}{\sigma_k^2}$	Each method gets a vote weighted by one over its miss squared. Income (miss 0.25) gets 16 votes, Grok's number (0.35) about 8, the category base rate (0.70) gets 2. The blend lands at \$570,768.
3. How sure is the blend	$\sigma^2 = \underbrace{\frac{1}{\sum_k w_k}}_{\text{each method's own miss}} + \underbrace{\frac{\sum_k w_k (\mu_k - \mu)^2}{\sum_k w_k}}_{\text{how far apart they land}}$	Two things go wrong: every method is a bit off, and the methods disagree. We add both. Squirrel Hill's three methods land between \$469k and \$589k, close together, so the total miss is 0.22.
4. What we show	$[V_{20}, V_{80}] = e^{\mu \mp 0.84 \sigma}, \quad \sigma \geq 0.12$	A range, never a bare number. 0.84 sigma is where the 20th and 80th percentiles sit on a bell curve: \$570,768 times 0.83 is \$475k, times 1.20 is \$686k. We never claim to be surer than 12 percent.

Where each miss comes from

Today's numbers come from BizBuySell's 2026 tables. We also have a check that takes real listings, hides the asking price, runs each method and records the miss. Once it has run on enough listings, measured misses replace this table.

Listing price	it is actually for sale	±15%
Income	owner earnings x category multiple	±25%
Grok's number	wider when K2 disagrees	±35%+
Proxy	revenue guessed from staff, machines, reviews	±60%
Base rate	median for the category in that state	±70%

Bloomfield Coin Laundry has only a Google pin, so two weak methods run: miss 0.82, range \$96k to \$384k. The page shows the wide range instead of hiding it.

The owner is the seller. There is no market maker.

The owner's asks. 3,000 shares (30 percent) in five lots of 600, at five prices just above our value. For Squirrel Hill, value \$57.08: lots at \$58.67, \$60.76, \$63.03, \$65.59 and \$68.62. A shakier valuation spreads those prices wider.

The owner's buyback. The owner also stands ready to buy up to 1,000 shares back at \$47.48, about 17 percent under our value. That bid is in the book from the first minute.

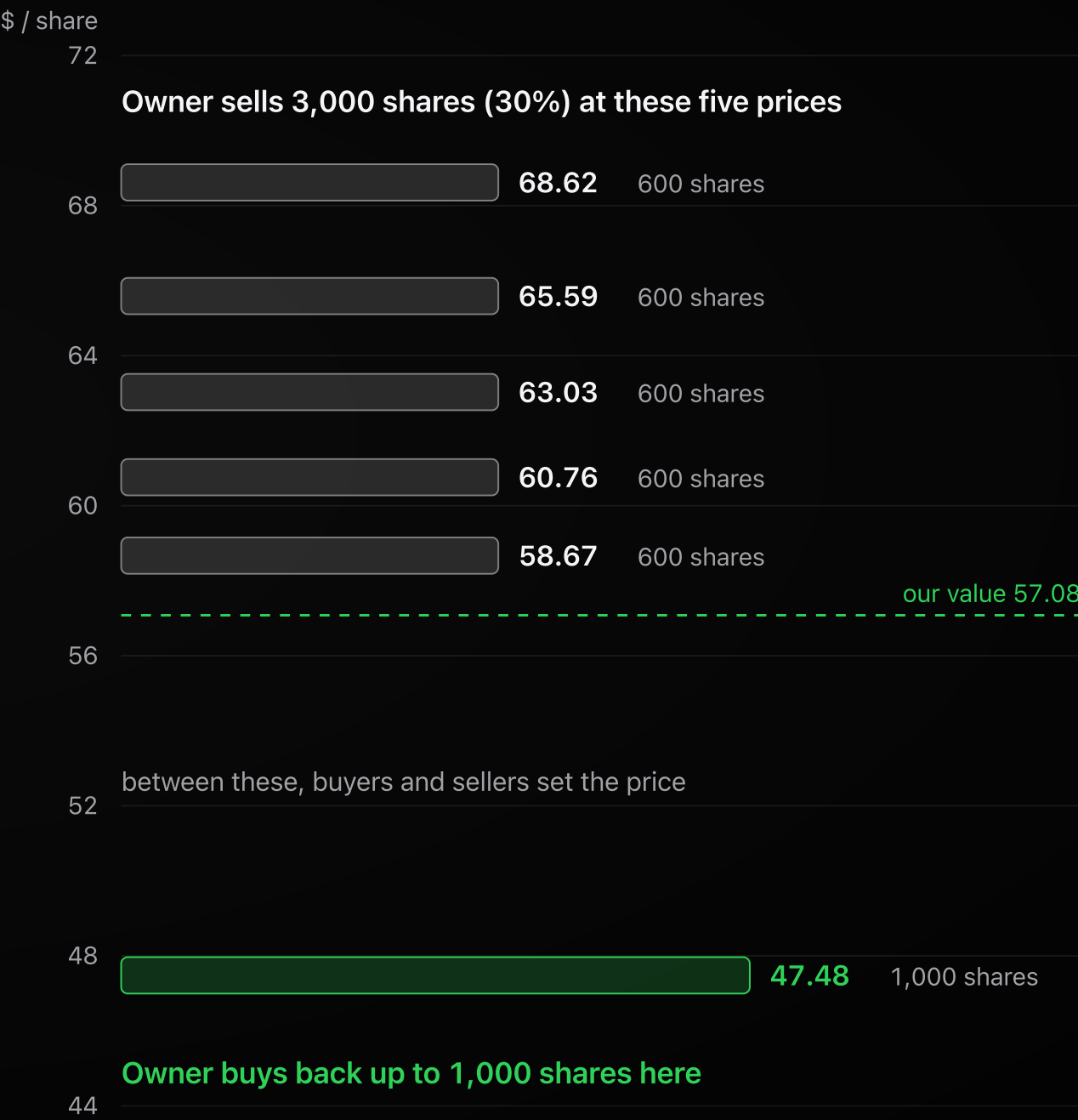
One price per round. Every 10 to 60 seconds, faster when the book is busy, all orders clear at once at a single price. Everyone in that round pays it. Between rounds our value updates from what buyers paid, and the owner's asks and buyback move with it. Bartr holds no shares and takes no side.

Which price does a round clear at?

$$p^* = \arg \max_p \min \left\{ \underbrace{D(p)}_{\text{shares buyers want at } p}, \underbrace{S(p)}_{\text{shares offered at } p} \right\}$$

Try every price. At each one, count how many shares buyers would take and how many sellers would give. Clear where the smaller of the two is largest. In round 1 of the demo, buyers wanted 305 shares at \$58.67 or above and the owner offered 600 there, so 305 traded at \$58.67, and Jonas, who bid \$64.20, paid \$58.67 too.

One price per round. The owner is on the other side.



Live, two rounds

Squirrel Hill Wash and Fold · 10,000 shares · our opening value \$57.08

RK

Rita K.
owner
sold 1,158.85 of 3,000

sell 100 @ 60.20

J

Jonas
Bloomfield
holds 288.85, sold 71.15

buy 150 @ 60.90

P

Priya
Oakland
holds 330

buy 200 @ 62.00

A

Ana
phone, in the room
holds 200

buy 120 @ 63.50

L

Leo
Shadyside
holds 190

buy 60 @ 54.80

T

Tom
Lawrenceville

buy 55 @ 54.60

K

Ken
Greenfield

buy 150 @ 61.50

S

Sam
phone, in the room
holds 150

buy 25 @ 49.10

N

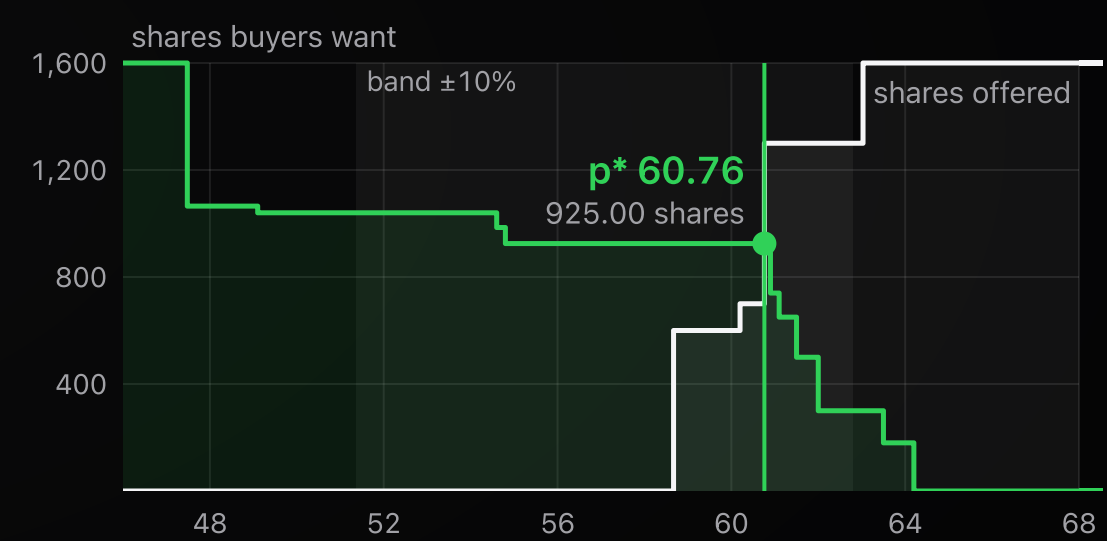
Nina
South Side

Bids		qty · limit	Asks		qty · limit
60.00	54.80	Tom	28.85	60.20	Jonas
55.00	54.60	Ken	173.08	61.35	owner
25.00	49.10	Nina	600.00	62.58	owner
1,000.00	53.65	owner floor	600.00	63.95	owner
			600.00	65.54	owner

After 2 rounds

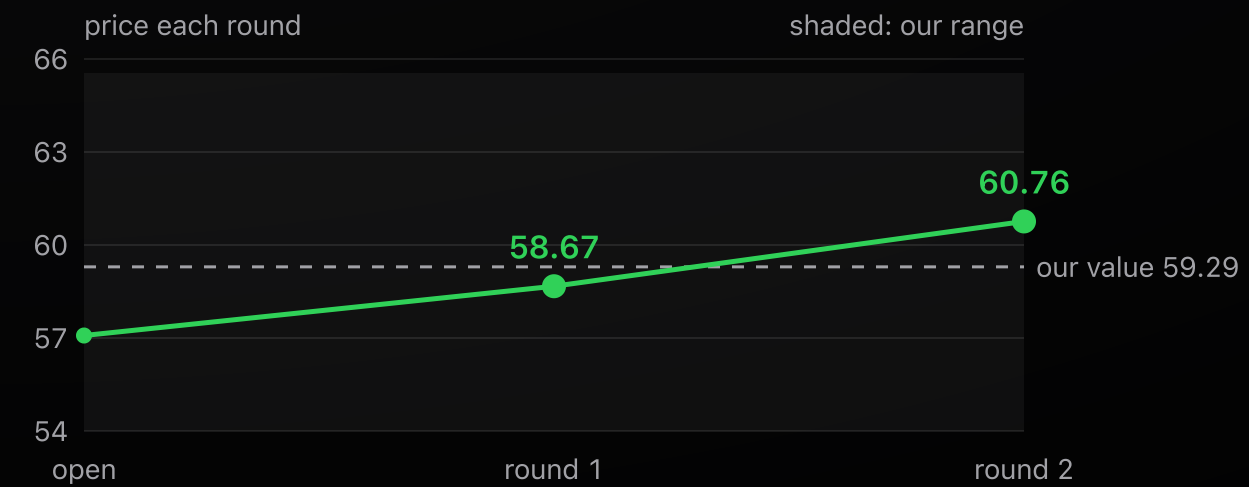
60.76

everyone paid this

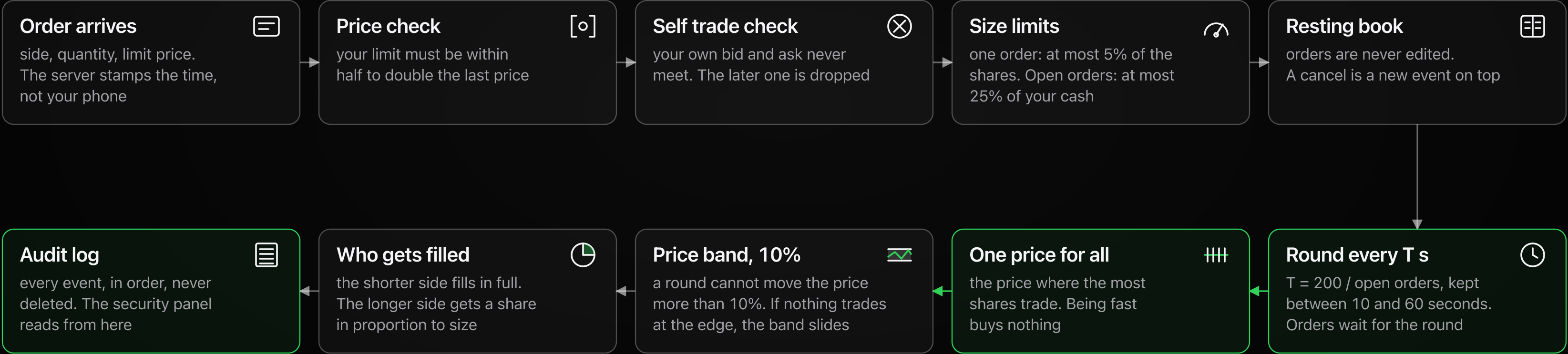


Round 2: 925.00 shares at 60.76. Jonas, Priya, Leo, Ana, Sam all paid 60.76, whatever they bid. 853.85 came from the owner. Jonas sold 71.15 of those to them, at the same price.

Buyers paid more than our 57.08, so our value moves to 59.29. The owner's asks and buyback bid move with it.



One price. No head start.



Nothing matches between rounds, so a faster connection gets you nothing. Two rounds in a row at the band edge halt the market for one round. The owner can only change quotes between rounds, never bids above its buyback price and never asks below its lowest ask, so the account holding 30 percent of the shares cannot chase the price. Every order carries a server timestamp and lands in the audit log.

Every suggestion shows the gap between our value and the market price. A slider decides how hard to lean on it.

The gap. Our value next to the last traded price. We only suggest businesses trading below our value; the gap is what you stand to gain when the price catches up.

How much to buy. The Kelly formula turns the gap and our uncertainty into a share of your cash. A big gap with a sure value gets more; a big gap with a shaky value gets less. No single business ever gets more than 20 percent.

Your risk level. A slider from 0.25 to 1.0 scales that share. Default is half, because our value is an estimate. Move it and the gain and the possible loss move with it.

How much of my cash should go in?

$$\text{gap} = \ln \frac{\text{our value}}{\text{price}}, \quad \text{share of cash} = \text{slider} \times \frac{\text{gap}}{\sigma^2}, \quad \text{capped at 20\%}$$

Iron City: value \$97.96, price \$94.50, gap 3.6 percent, miss 0.36. Slider at 0.5: 0.5 x 0.036 / 0.13 = 14 percent of your cash.

Last traded price

\$94.50

Our value

\$97.96 miss 0.36

Gap, what you stand to gain

+3.6%

Your risk level

0.50

0.25 cautious

1.0 full Kelly

Suggested stake

\$13,900

Gain if the price reaches our value

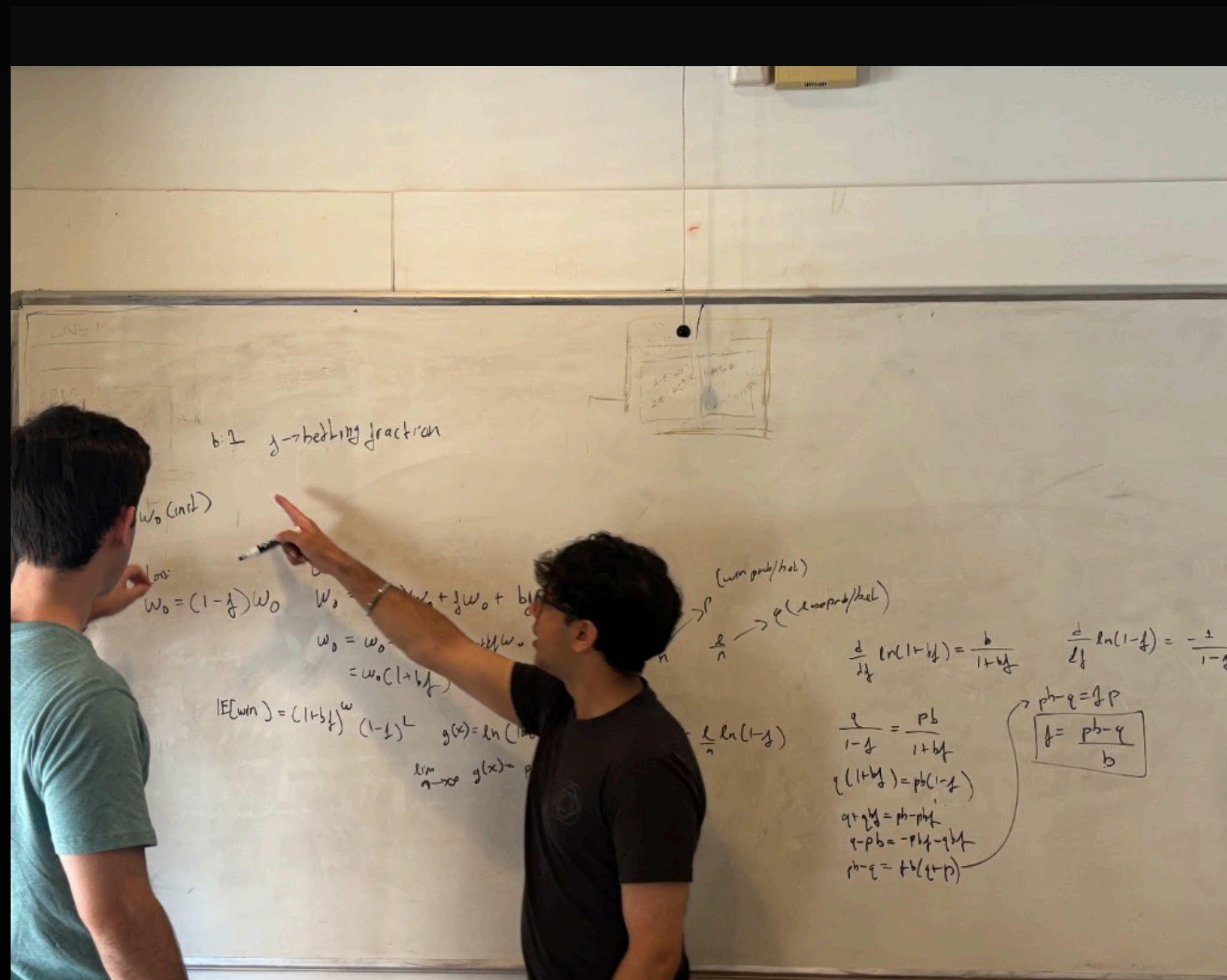
+\$508

Loss if our value is one miss too high

-\$4,194

cash \$100,000 · stake = slider x 0.28 x cash, capped at \$20,000

We backed the bidding price and the portfolio construction with rigorous quantitative math.



Kelly Criterion Derivation:

$w_0 \rightarrow$ init capital $b \rightarrow$ percent return (excluding init)
(return if we win)

$f \rightarrow$ optimal fraction to bet

If won: $w_1 = (1-f)w_0 + f w_0 + f b w_0$ If lost: $w_1 = (1-f)w_0$
 $= w_0 - f w_0 + f w_0 + f b w_0$
 $= w_0(1-f)$

$w = \# \text{ wins}$ $L = \# \text{ losses}$

$$w_n = w_0(1+bf)^w (1-f)^L$$

$$\ln\left(\frac{w_n}{w_0}\right) = w \ln(1+bf) + L \ln(1-f)$$

$\uparrow$ growth of wealth
 $\lim_{n \rightarrow \infty} \ln\left(\frac{w_n}{w_0}\right) = \frac{w}{n} \ln(1+bf) + \frac{L}{n} \ln(1-f)$

$$g(f) = p \ln(1+bf) + q \ln(1-f)$$

maximize $g(f)$:

$$0 = \frac{pb}{1+bf} - \frac{q}{1-f} \Rightarrow \frac{q}{1-f} = \frac{pb}{1+bf}$$

$$\Rightarrow q + qbf = pb - pb f \Rightarrow pb - q = f b (q + p)$$

$$\Rightarrow f = \frac{pb - q}{b} \quad \square$$

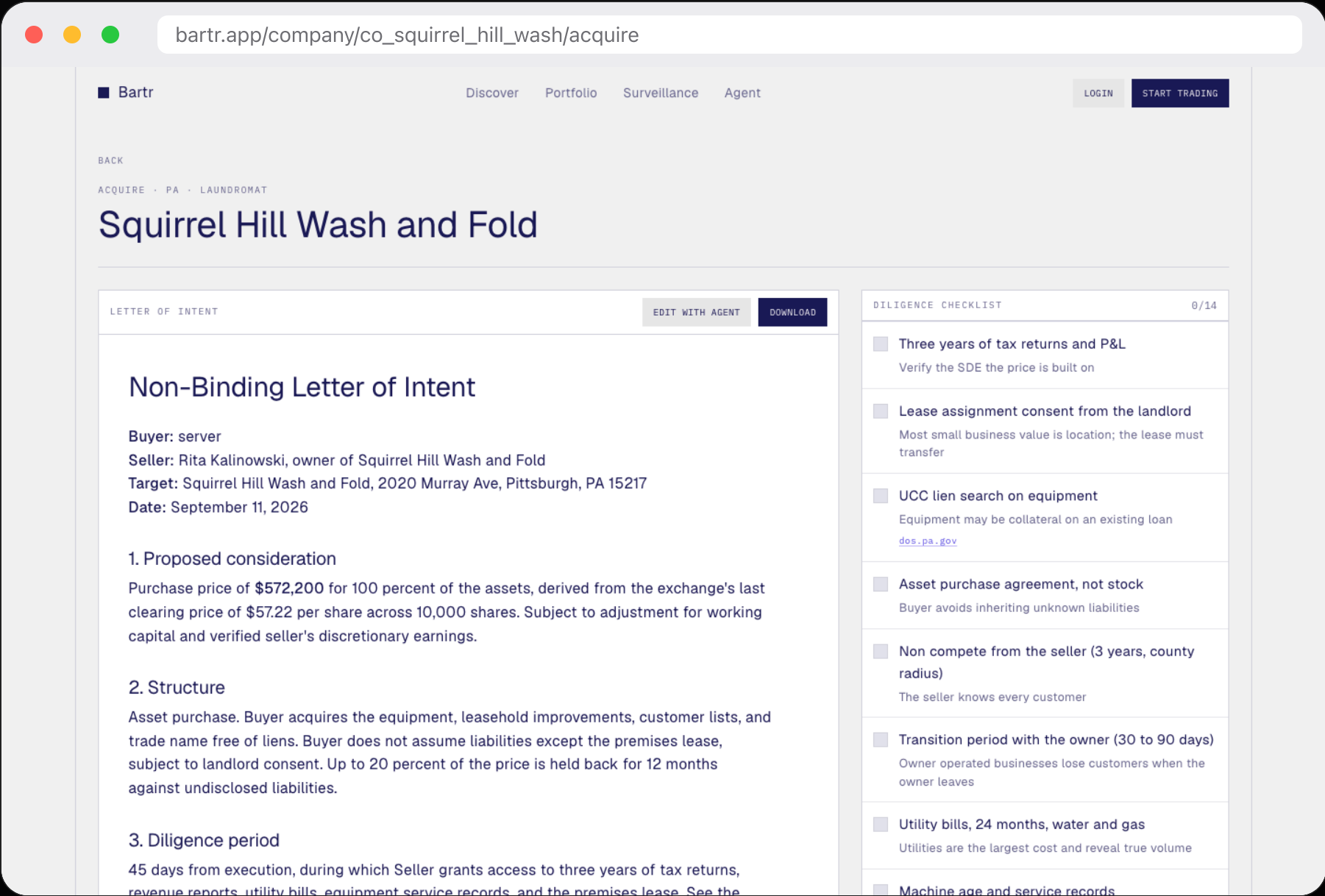
From 20 shares to the whole company.

Bartr drafts two things for you:

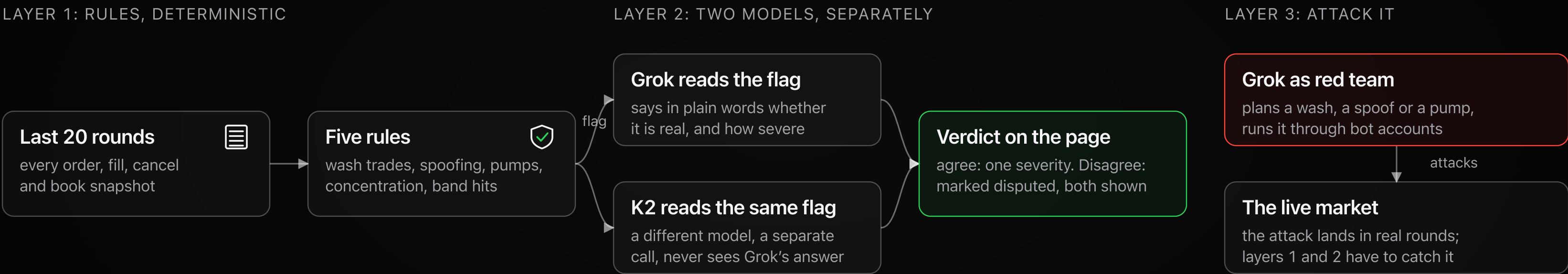
A letter of intent priced at the last traded price.

A diligence checklist for a Pittsburgh business: city registration, Allegheny County health permit, PA bulk sale clearance, lease, liens, tax returns.

Each government item links to the official page.



Rules catch it. Two models judge it. Grok tries to beat it.



● **Flagged** wash trading, high

Rule: wash_a and wash_b have traded 9 times and only with each other. The volume looks real but neither one's stake ever changed.

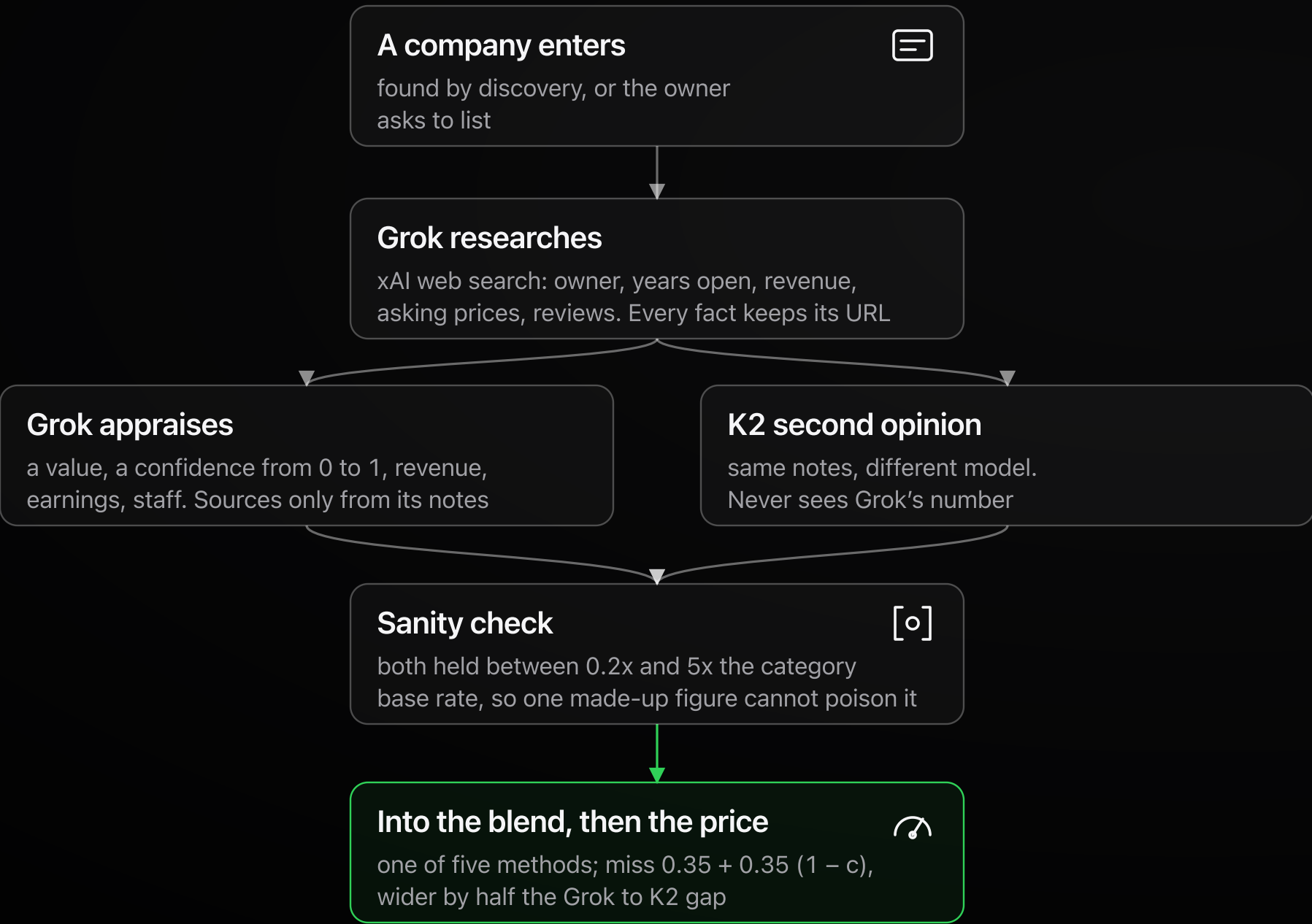
Grok high · K2 high · agree

Grok (example): "Alternating sides at the clearing price with no other counterparties. No economic purpose; treat as manipulation." K2 reached the same severity without seeing it.

What happens next

The flag and both reviews go on the admin dashboard and into the audit log. Two rounds in a row at the price band halt a market. Every red team attempt is logged with whether it was caught.

Every company that enters the exchange is appraised twice: Grok researches it and names a value, K2 names one without seeing Grok's, and both go into the price.



Everywhere else Grok runs

Reading the web. Builds a profile from Querit pages. A number is kept only if the exact sentence it came from is on the page, and financials must be in dollars with a date.

Security. Every rule flag goes to Grok and to K2 in separate calls. As red team, Grok plans an attack and runs it through bot accounts.

Chat. An agent with tools: search, open a company, read the book, place an order, get suggestions. The same agent answers over iMessage. Also an owner persona per company, a market narrator, and a market health memo.

Portfolio and acquire. Reads your profile into limits, writes the reason under each suggestion, drafts the letter of intent and the diligence checklist.

When Grok is down. Every call has a plain code fallback. Calls through the shared wrapper are cached and listed on /readiness.

How it runs

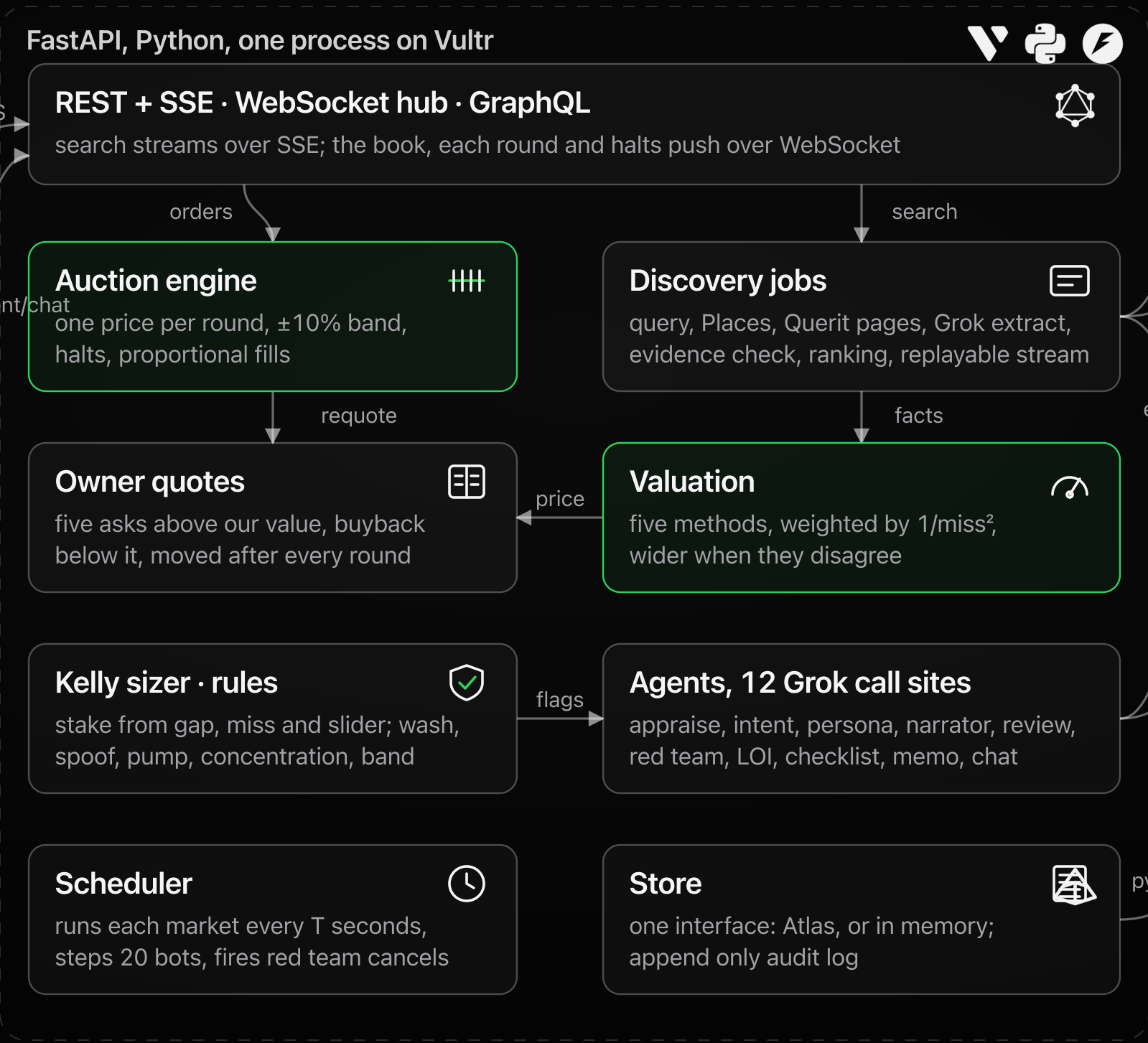
103 tests pass offline; every model call has a fallback

CLIENTS

Web app
Next.js 16 on Vercel. Discover, company, portfolio, acquire, surveillance, chat

iMessage
Photon bridge in Bun; texts go to the same chat agent

API



DATA AND MODELS

Google Places
businesses, ratings, review counts

Querit
reads the pages about each business

xAI · Grok
grok-4.6 with web search for research; a fast model for chat and reviews

IFM · K2 Horizon
second opinion on every appraisal and every flag

MongoDB Atlas
companies, markets, orders, rounds, trades, audit; migrations

HTTPS

agent/chat

orders

search

requote

price

facts

extract

flags

pymongo

Bartr.

Nobody could invest in the laundromat on Murray Avenue until tonight. Buy 20 shares and watch the next round clear.

Grok
xAI

IFM K2
Horizon

Querit
web search

 **MongoDB Atlas**
database

 **Auth0**
login

 **Vultr**
API host

 **Vercel**
web host

 **Cursor**
editor



1

Kelly Criterion Derivation:

$w_0 \rightarrow$ init capital $b \rightarrow$ percent return (excluding init)
(return if we win)

$f \rightarrow$ optimal fraction to bet

$$\begin{aligned}
 \text{If won: } w_1 &= (1-f)w_0 + fw_0 + fbw_0 & \text{If lost: } \underline{w_1} &= \underline{(1-f)w_0} \\
 &= w_0 - fw_0 + fw_0 + fbw_0 \\
 &= \underline{w_0(1-f)}
 \end{aligned}$$

$w = \# \text{ wins}$ $L = \# \text{ losses}$

Appendix · The clearing rule and the belief update

One round clears where the most shares trade

$$D(p) = \sum_{b: \ell_b \geq p} q_b, \quad S(p) = \sum_{s: \ell_s \leq p} q_s, \quad p^\star = \arg \max_{p \in \mathcal{L}} \min\{D(p), S(p)\}$$

ties: $\arg \min |D - S|$, then nearest p_{t-1} ; $p^\star \leftarrow \text{clip}(p^\star, (1 - \beta)p_{t-1}, (1 + \beta)p_{t-1}), \beta = 0.10;$ $x_i = q_i \frac{\min\{D, S\}}{\max\{D, S\}}$ on the long side

Our value moves with what people paid

$$\ln V \sim \mathcal{N}(\mu, \sigma^2) \text{ (prior from five estimators),} \quad \mu_{\text{post}} = \frac{\mu/\sigma^2 + \sum_t w_t \ln(Np_t^\star)/s_m^2}{1/\sigma^2 + \sum_t w_t/s_m^2}, \quad \sigma_{\text{post}}^2 = \frac{1}{1/\sigma^2 + \sum_t w_t/s_m^2}, \quad w_t = \frac{v_t}{\bar{v}}$$

The owner quotes from the same distribution

$$\text{floor bid} = \frac{V_{0.20}}{N}, \quad \text{ask ladder} = \left\{ \frac{V_q}{N} : q \in \{0.55, 0.6125, 0.675, 0.7375, 0.80\} \right\}, \quad V_q = \exp(\mu + \sigma \Phi^{-1}(q))$$

The stake

$$\text{gap} = \ln \frac{v}{p}, \quad f^\star = \frac{\text{gap}}{\sigma^2}, \quad f = \text{clip}\left(\frac{1}{2}f^\star, 0, 0.20\right), \quad \$ = f \cdot B$$

D demand, S supply, ℓ limit, q quantity, N = 10,000 shares, p_{t-1} last clearing price, v_t round volume, s_m per round price noise, Φ⁻¹ inverse normal CDF, B bankroll. Same constants as engine.py.